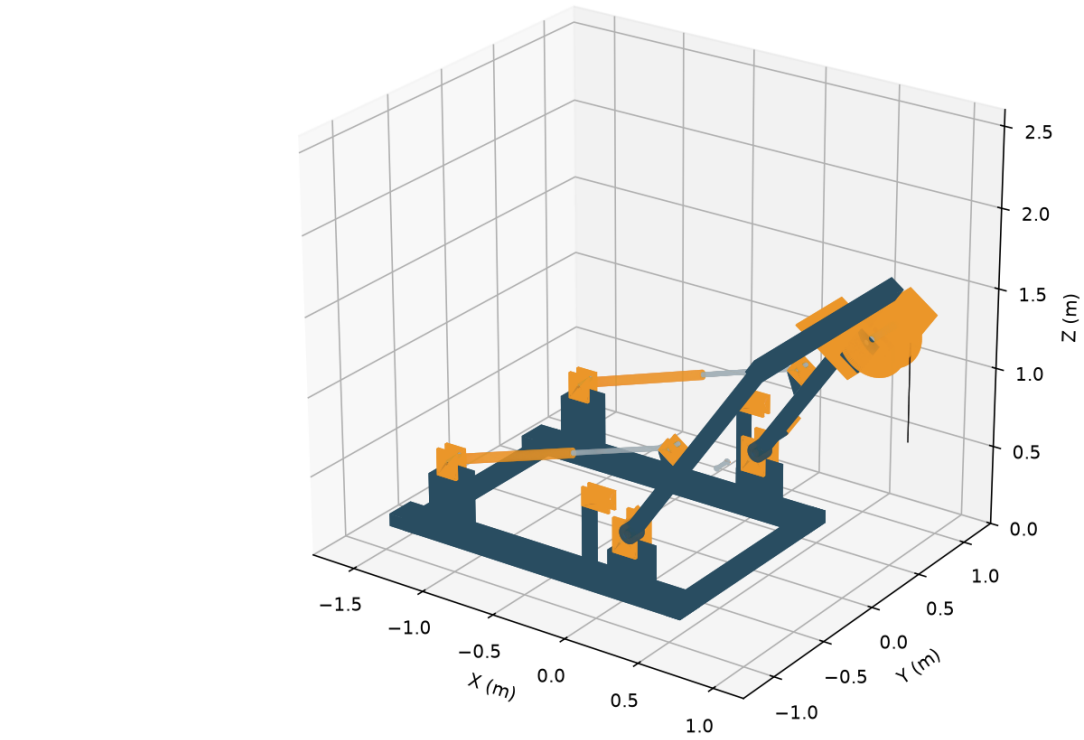
ROV launch-and-recovery frame

Final portfolio concept | geometry, mechanisms and reproducible engineering screens



Purpose

Explore a 100 kg assumed ROV handling system with a pivoting frame and twin hydraulic actuators.

Final model

80 valid solids: structural members, retained pins, parking locks, drum, bearing supports and cylinder envelopes.

Design decisions

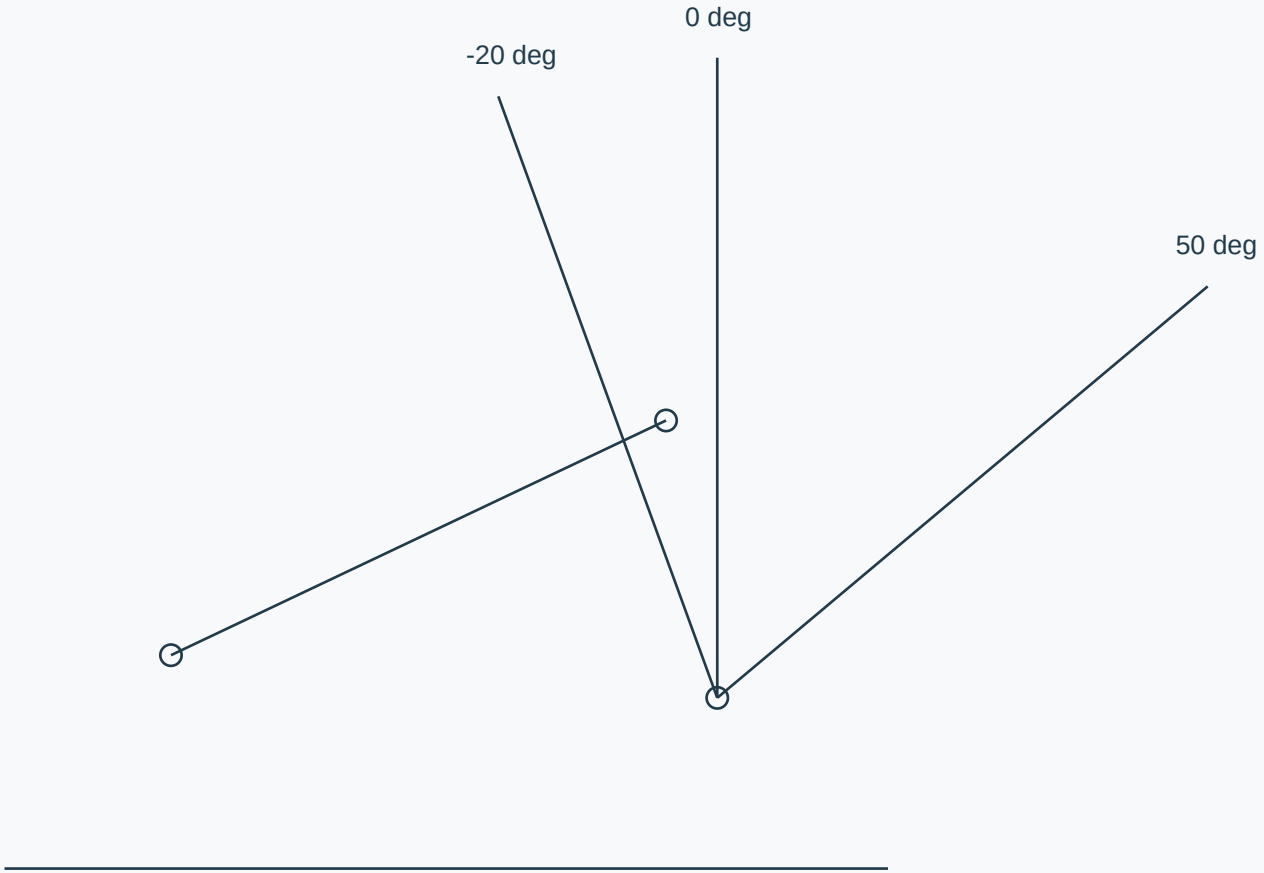
Rearward actuator attachments improve clearance. Direct drum payout avoids a short fixed rope lead. Pins restrain the unloaded parked frame.

Scope

A completed student concept study, not a certified lifting appliance. Parked position is not a fully folded transport configuration.

D-01 | Layout and actuator envelope

Side-view construction diagram | dimensions in mm | not to scale



Main pivot: (0, 400)

Fixed actuator pin: (-1280, 500)

Upright moving pin: (-120, 1050)

Pivot-to-head: 1500

Leg centres: 1300

Legs: 80 x 80 x 5 SHS

Head: 100 x 100 x 6 SHS

Reference cylinder: ABCo NFR2300800. Bore 50, rod 30, stroke 800; pin centres 1000 closed / 1800 open. Eye bore 25.25; eye width 45.

Required pin centres: 1055.3 to 1749.5; travel 694.2. End margins: 55.3 retracted and 50.5 extended. Reference supplier temperature limit: -25 C. Marine suitability, stroke tolerances and low-temperature approval are not established.

D-02 | Joints, parking and rope path

Interface schedule and functional arrangement | nominal dimensions

Retained joints

Main pivot: 40 mm pin, 80 mm bush length, 140 mm clevis plate centres and 20 mm plates. Actuator pins: 25 mm; base plates 20 mm at 80 mm centres; moving plates 15 mm at 70 mm centres. CAD adds flanged heads, washers and cross-drilled keeper envelopes. Fits and keeper strength remain preliminary.

Unloaded parking restraint

At -20 degrees, two 25 mm removable pins engage outboard tabs and fixed forks, 240 mm from the pivot. Operating CAD stores the pins on the skid; parked_locked.step shows engagement. An illustrative 2g frame-only case gives 2.10 kN per pin. This is not a sea-fastening rating.

Direct payout drum

400 mm rope pitch diameter; 140 mm face; assumed 6 mm rope and 7.5 mm lay pitch. Eighteen whole turns minus three retained turns gives 18.85 m usable single-layer capacity, against 15 m study payout. Drum grooves and wound rope are not modeled.

Hoist requirements

At the 2.0 payload multiplier: 1.962 kN line pull and 392.4 Nm ideal drum torque before losses. Motor, brake, rope construction, minimum breaking load and terminations require product selection. The motor/brake mass is an assumed 12 kg.

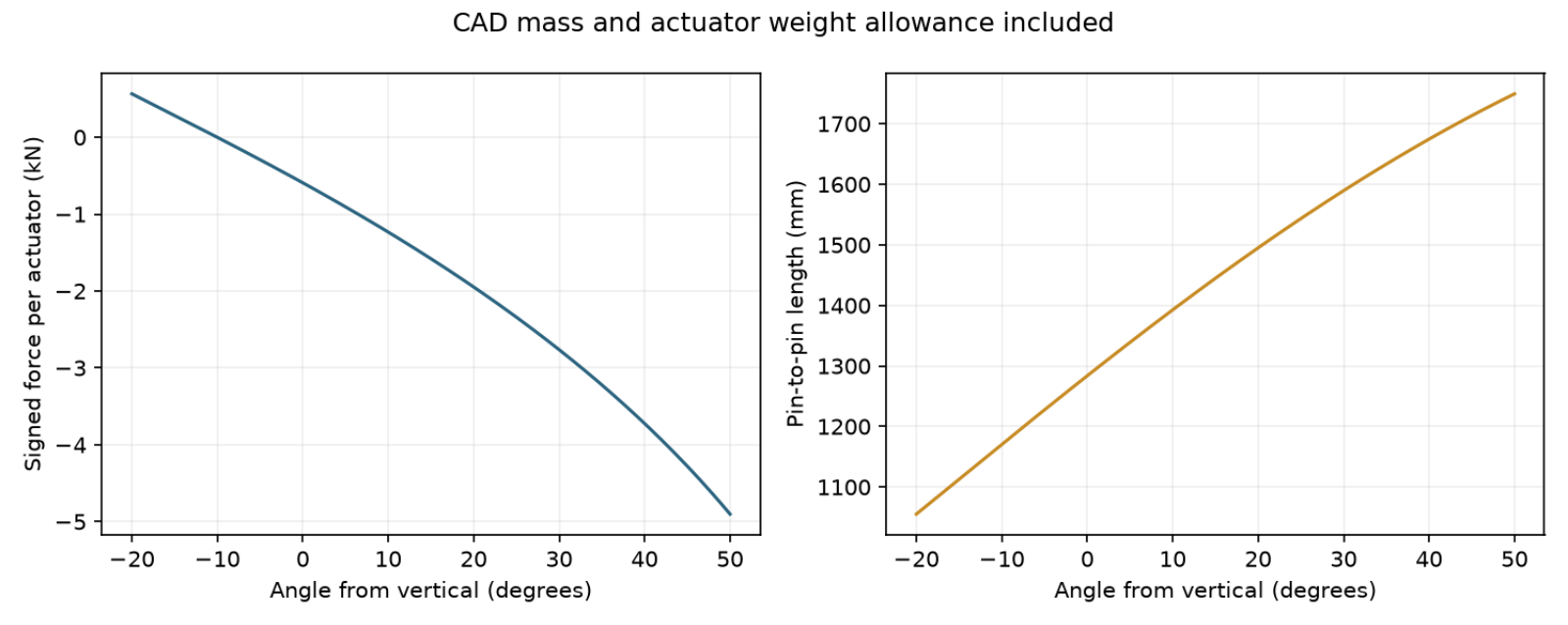
Cable management

The lifting line drops vertically from the overhung drum, with a +/-70 mm lateral payout envelope. The ROV electrical tether is separate. Hose loops, moving payload swing, spooling control and tether routing are outside the modeled envelope.

C-01 | Motion, gravity and force

Static moment balance with virtual-work cross-check | 141 sampled positions

Moving CAD group: 145.7 kg. Local CG: X 76.9, Y 22.2, Z 1155.0 mm. Two 13 kg cylinders contribute a 13 kg moving-equivalent gravity allowance. Payload: 100 kg.



Payload factor 1.0; side share 50%: actuator 3.85 kN; pivot reaction 4.33 kN.

Payload factor 1.5; side share 50%: actuator 4.90 kN; pivot reaction 5.46 kN.

Payload factor 2.0; side share 60%: actuator 7.14 kN; pivot reaction 7.91 kN.

These factors are sensitivity assumptions, not class-rule load cases. Signed actuator force distinguishes push and pull; both directions need controlled load holding. No vessel acceleration, snag or lateral loading is validated.

C-02 | Screening results and verification

Simplified models support design comparison; they do not establish rated capacity

25 mm actuator pin: combined simple bending/shear	93.99 MPa
40 mm pivot pin: combined simple bending/shear	44.39 MPa
Actuator lug: projected bearing	9.53 MPa
Upper leg: bending-only screen	76.56 MPa
Head: bending-only screen	16.52 MPa

Calculation basis

Higher-load case uses payload factor 2.0 and 60% side share. Pin shear F/(2A), bending moment F*s/4, and bearing F/(2td). Nominal cylinder force at 100 bar: 19.63 kN push / 12.57 kN pull before losses. Ideal 30 mm rod Euler load at 1900 mm pinned length: 21.74 kN; supplier buckling verification is still required.

CAD checks

80 valid solids. Selected interference checks at 15 positions found no overlaps, including three sampled rope payout lanes. Minimum selected frame/foundation gap: 12.5 mm; cylinder/leg: 82.2 mm. This is discrete selected-pair checking, not a complete swept-envelope assessment.

Unrated interfaces

No certified material grade, fatigue life, weld capacity, deck foundation, hydraulic circuit or brake rating. Combined frame stresses and local bracket details need further analysis. No physical testing or FEA was performed.

R-01 | Design record and handoff

What the portfolio demonstrates, what the next engineering stage must resolve

Operating concept

Park unloaded and engage both locking pins. Before motion, withdraw both pins and confirm clearance. A future control design needs pin-status and travel interlocks, synchronized cylinders, bidirectional load holding, controlled lowering and emergency stop. This package is not an operating procedure for real equipment.

Portfolio explanation

I developed an AI-assisted parametric ROV handling concept, compared actuator geometry, incorporated CAD mass into static loads, and designed retained joints and an unloaded parking restraint. I checked selected clearances across the motion range and documented supplier dimensional limits. These are calculated concept results, not tested lifting capacity.

Package contents

Editable CadQuery source, STEP assembly and three motion poses, parked-lock assembly, mass/BOM CSV, actuator sweep, JSON calculation results, rendered views, motion schematic and this report.

Primary references

ABCo / Steerforth NFR2300800 product and stock-cylinder dimension sheet: cylinder geometry, mass and temperature range. Parker Designing With Cylinders: mounting and buckling context. Ingersoll Rand fleet-angle guidance: fixed lead considerations. Links and access notes are in REFERENCES.txt.

ABCo cylinder reference

Published cylinder dimensions

Parker cylinder design guidance

Ingersoll Rand fleet-angle guidance